Session 2: Spatial foundations, data acquisition and quality

Modeling the distribution of biodiversity under global climate change

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Principles of Geographic Information Systems (GIS)

What is GIS? Why does it matter for modeling biodiversity?

Definition: a system to capture, store, manipulate, analyze, and present spatial or data.

Relevance in macroecology: mapping habitats, tracking species distributions under climate change, analyzing environmental patterns (e.g., temperature), conservation planning (MPAs).

Specific link to SDMs: supports the representation of species locations and environmental variables; crucial for data preparation, interpretation and visualization.

Representing Reality: Vector Data Models

Represents **discrete features with distinct boundaries**.

Primitives:

Points: x, y (,z) locations (e.g., occurrences, stations).

Lines: points connected by segments (e.g., coastlines, tracks, currents).

Polygons: closed sequence of segments representing areas (e.g., MPAs, islands).

Features are linked to an attribute table (e.g., species name, MPA size).

Can store spatial relationships (e.g., distances between sites).



Spatial information for *Caretta caretta*

Locations

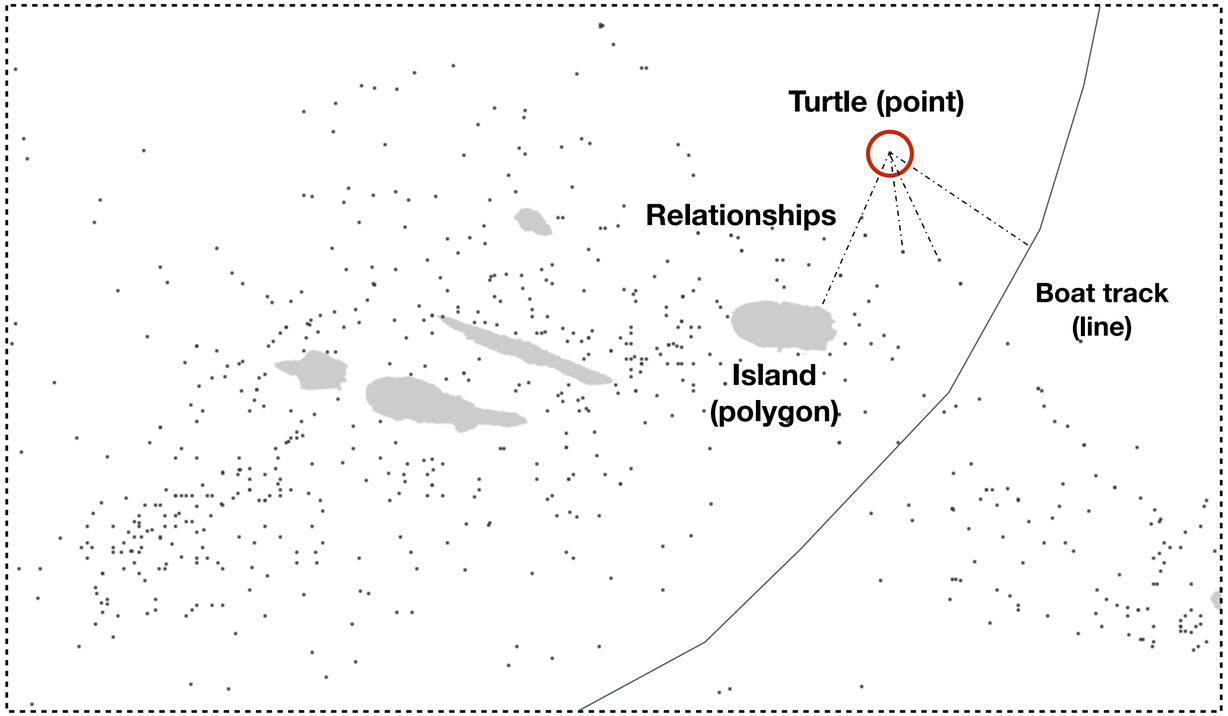
Attributes

Relationships

Lon	Lat	Species	Common name	Date	Depth	Behaviour	Observer	Distance to land
0,61287	39,58955	Caretta caretta	Loggerhead sea turtle	10/08/12	22	Swimming	John Doe	232
0,83999	39,77417	Caretta caretta	Loggerhead sea turtle	03/03/12	12	Swimming	John Doe	3454
3,18192	42,40479	Caretta caretta	Loggerhead sea turtle	14/08/11	1	Eating	Unknown	23
-4,86395	55,24295	Caretta caretta	Loggerhead sea turtle	01/01/04	0	Swimming	Unknown	2567
-5,89297	57,06977	Caretta caretta	Loggerhead sea turtle	03/03/12	0	Sleeping	John Doe	878
34,9082	32,55822	Caretta caretta	Loggerhead sea turtle	22/12/02	23	Sleeping	John Doe	777
0,61287	39,58955	Caretta caretta	Loggerhead sea turtle	14/08/11	12	Swimming	John Doe	6
0,83999	39,77417	Caretta caretta	Loggerhead sea turtle	03/03/12	2	Sleeping	John Doe	78

Vector data model

Points, lines and polygons (locations, and features with attributes)



The distribution of the Loggerhead sea turtle in the Azores Islands

Common file formats

Vector data are common as **shapefiles** (a standard since 1990).

.shp is the geometry of vector features

.dbf are the attributes of vector features

.shx helps the GIS Application to find features more quickly.

Vector data are migrating to simpler formats like **GeoPackage files**, which are an open, portable, self-describing, compact formats for spatial information.

.gpkg is the geometry and the attributes of vector features.

saccorhiza

ID	SPECIES	Y	LON	LAT	REF
1	Sacchorhiza	polyschides	1966	-5.600819	36.018775	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula Ibèrica e Islas Baleares. IX
2	Sacchorhiza	polyschides	1993	-5.601822	36.019239	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula Ibèrica e Islas Baleares. IX
3	Sacchorhiza	polyschides	1960	-5.653864	36.062617	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula Ibèrica e Islas Baleares. IX
4	Sacchorhiza	polyschides	1965	-5.44705	36.116542	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula Ibèrica e Islas Baleares. IX
5	Sacchorhiza	polyschides	1958	-5.353586	36.14075	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula Ibèrica e Islas Baleares. IX
6	Sacchorhiza	polyschides	1965	-5.922481	36.190028	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula Ibèrica e Islas Baleares. IX
7	Sacchorhiza	polyschides	1966	-6.087778	36.277683	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula Ibèrica e Islas Baleares. IX
8	Sacchorhiza	polyschides	1983	-5.158386	36.422061	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula Ibèrica e Islas Baleares. IX
9	Sacchorhiza	polyschides	1982	-4.717386	36.493706	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula Ibèrica e Islas Baleares. IX
10	Sacchorhiza	polyschides	1982	-4.882289	36.510117	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula Ibèrica e Islas Baleares. IX
11	Sacchorhiza	polyschides	1982	-4.622617	36.571558	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula Ibèrica e Islas Baleares. IX
12	Sacchorhiza	polyschides	1966	-4.421261	36.721283	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula Ibèrica e Islas Baleares. IX
13	Sacchorhiza	polyschides	1970	-8.936489	37.009864	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula Ibèrica e Islas Baleares. IX
14	Sacchorhiza	polyschides	1963	-8.978044	37.023033	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula Ibèrica e Islas Baleares. IX
15	Sacchorhiza	polyschides	1963	-8.667864	37.079811	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula Ibèrica e Islas Baleares. IX
16	Sacchorhiza	polyschides	1963	-8.266667	37.083333	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula Ibèrica e Islas Baleares. IX
17	Sacchorhiza	polyschides	1956	-8.45	37.083333	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula Ibèrica e Islas Baleares. IX. Saccor
18	Sacchorhiza	polyschides	1970	-8.247875	37.089072	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula Ibèrica e Islas Baleares. IX
19	Sacchorhiza	polyschides	1970	-8.469325	37.098511	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula Ibèrica e Islas Baleares. IX
20	Sacchorhiza	polyschides	1970	-8.896	37.184133	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula Ibèrica e Islas Baleares. IX. Saccor
21	Sacchorhiza	polyschides	1954	-8.770147	37.757611	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula Ibèrica e Islas Baleares. IX
22	Sacchorhiza	polyschides	1970	-8.796936	37.906147	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula Ibèrica e Islas Baleares. IX
23	Sacchorhiza	polyschides	1970	-9.216664	38.416647	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula Ibèrica e Islas Baleares. IX
24	Sacchorhiza	polyschides	1979	-9.1176	38.43685	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula Ibèrica e Islas Baleares. IX. Saccor
25	Sacchorhiza	polyschides	1970	-9.101489	38.444469	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula Ibèrica e Islas Baleares. IX
26	Sacchorhiza	polyschides	1989	-8.984847	38.469364	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula Ibèrica e Islas Baleares. IX
27	Sacchorhiza	polyschides	1982	-8.990794	38.495931	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula Ibèrica e Islas Baleares. IX
28	Sacchorhiza	polyschides	1970	-9.355975	38.689786	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula Ibèrica e Islas Baleares. IX
29	Sacchorhiza	polyschides	1989	-9.483333	38.7	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula Ibèrica e Islas Baleares. IX. Saccor
30	Sacchorhiza	polyschides	1951	-9.269286	38.702911	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula Ibèrica e Islas Baleares. IX

Common file formats

Vector data are also available as **text delimited files** (e.g., .txt, .csv), delimited by a **Tab, a comma or semicolon**. Useful for point data and matrices (e.g., excel).

Representing Reality: Raster Data Models

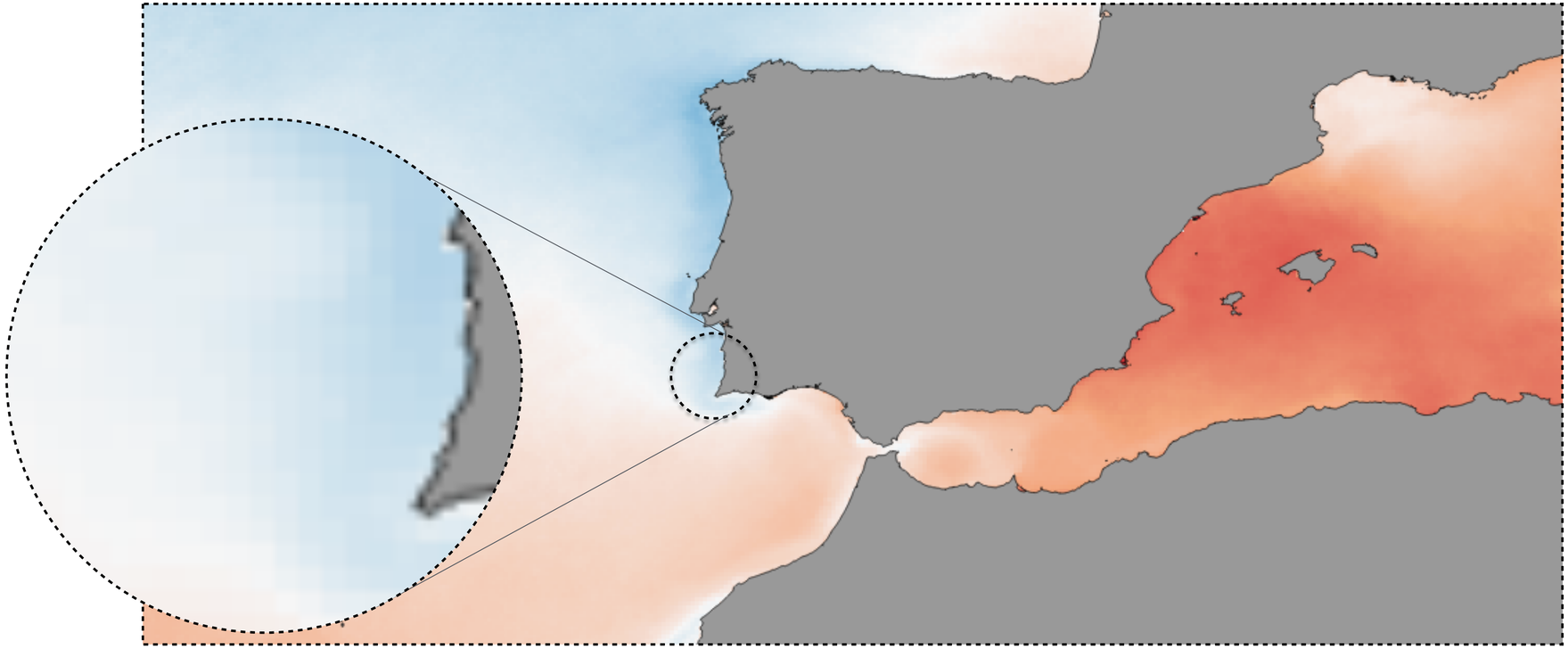
Composed of a **grid of cells** (pixels), where each cell **contains a value representing a specific feature** (e.g., temperature, depth, suitability score).

Raster data **can be continuous** (e.g., temperature) or **thematic** / categorical (e.g., habitat type).

SDMs rely heavily on digital raster layers to link biological occurrence data to extant environmental conditions

Raster data model

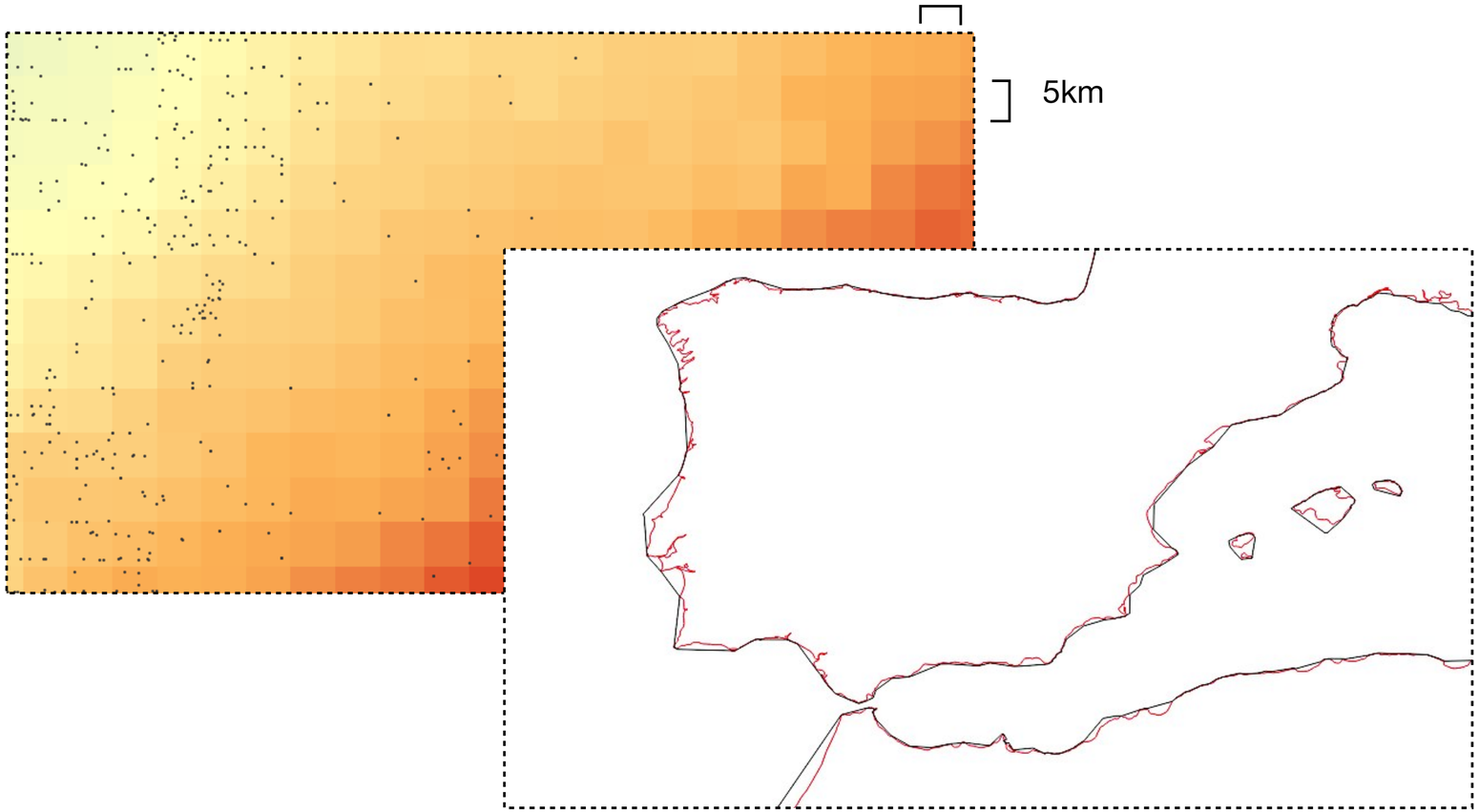
Grids and cells representing continuous features



SST in the western Mediterranean Sea and adjacent Atlantic Ocean.

Common file formats

Raster data are the the industry standard for GIS and satellite remote sensing applications. **Files are typically as .tif, .asc or .nc**.



Both models have well-defined spatial **extent and resolution**, which is the size of the **smallest feature able to be recognized.**

Vector data model: number of vertex;

Raster data model: size of cells;

Where on earth? Coordinate Reference Systems (CRS)

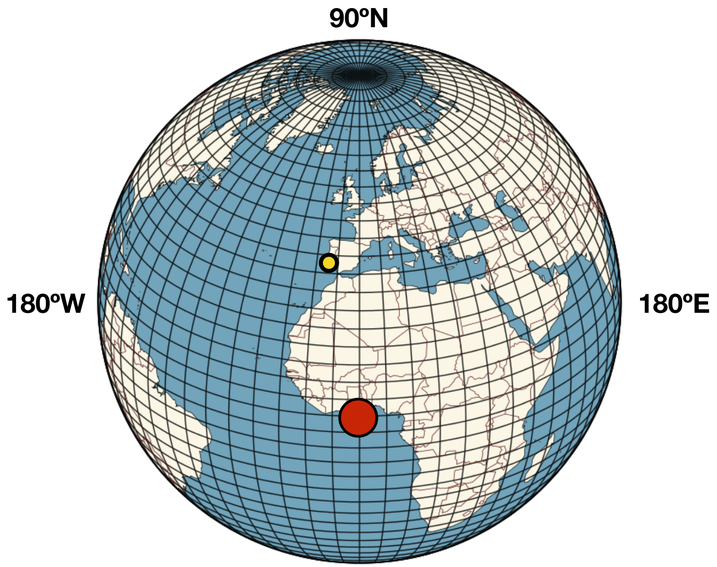
Coordinate reference systems **allow representing the locations of features within a common geographic framework**.

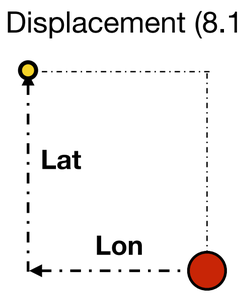
Without a CRS, coordinates are numbers that do not represent a specific place.

e.g., 7.321 ; -121.981

A CRS gives meaning to the information of location.

CRS have an inherent **fundamental challenge**: representing the 3D Earth on 2D surfaces.





Reference point (Datum) at the interception between:

Equator & Prime Meridian (0°N 0°E)

CRS type 1: Geographic Coordinate Systems (GCS)

Defines locations directly on the 3D Earth using latitude and longitude (angles).

Components:

- **Datum:** positions locations relative to the Earth.
- **Displacement units:** degrees (usually decimal degrees).

Decimal degrees are simpler to use.

37°2'38.08"N ; 7°58’20.73”W

37.043911° ; -7.972425° 👉🏽

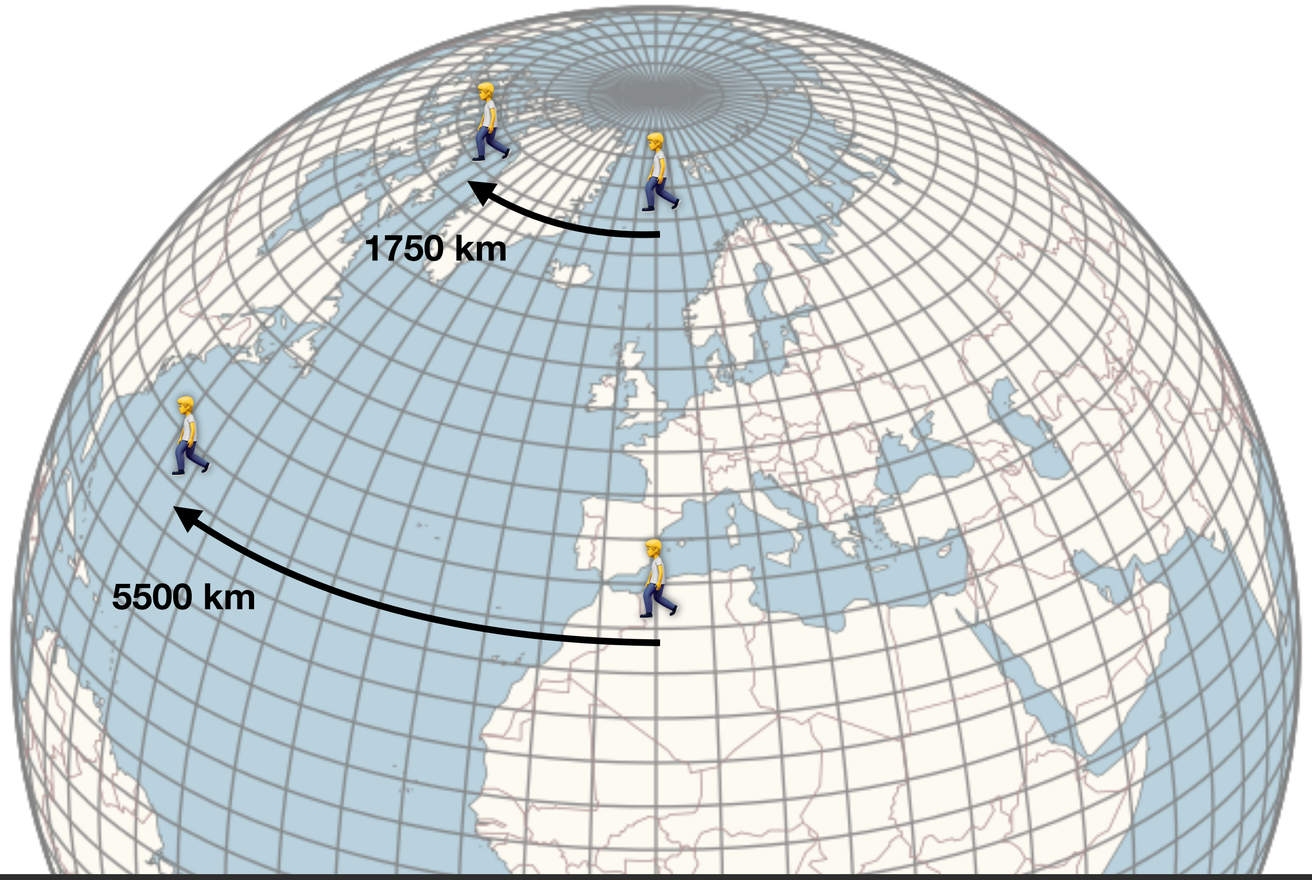
Simpler to type and share with others;

In data tables (e.g., R, Excel), longitude and latitude entries **use single cells, allowing direct analyses** (e.g., calculate distances).

Geographic coordinate systems

GCS **are not good to make measurements**.

There is the need to project data to an alternative coordinate system (e.g., based on actual distances).

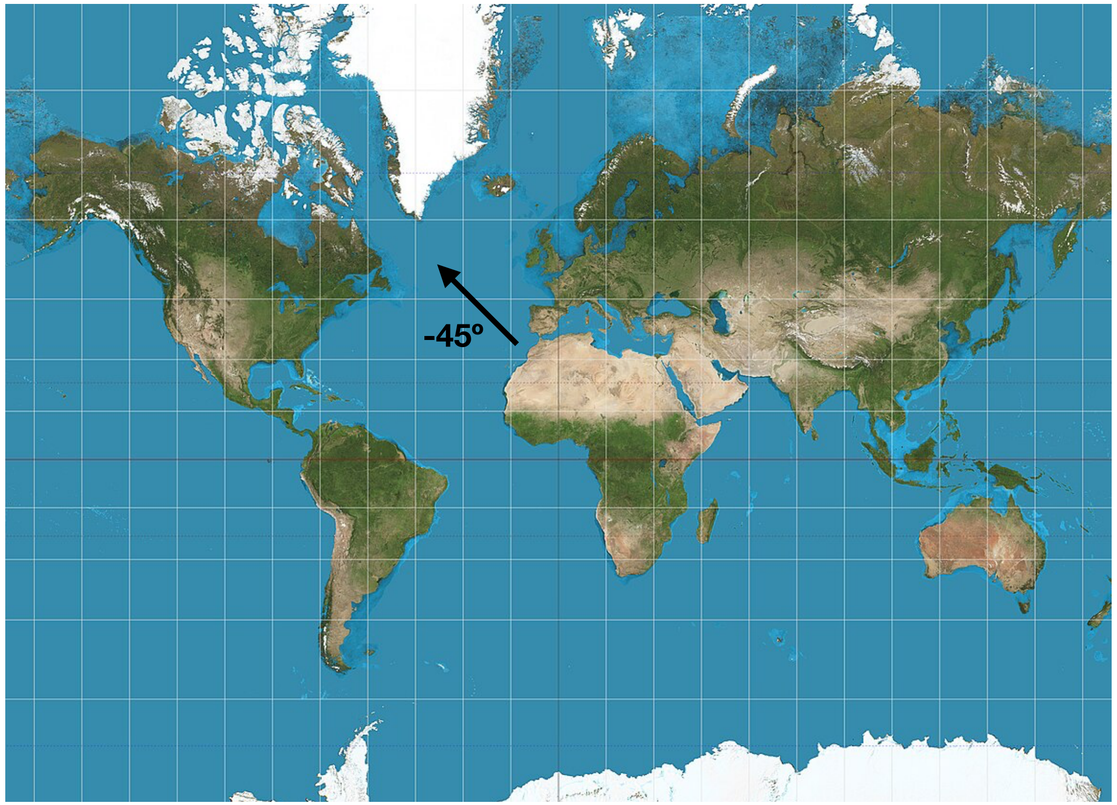


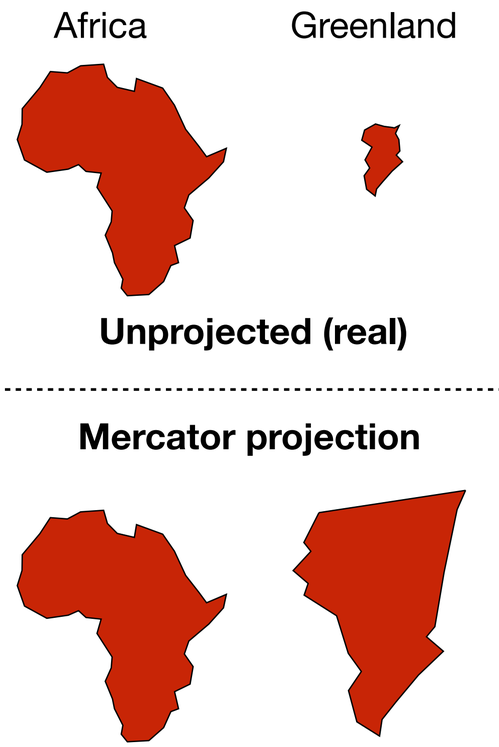
Displacement of 60º performed at different latitudes (360º longitude).

CRS type 2: Projected Coordinate Systems (PCS)

Mathematical transformation (**projection**) of the Earth's surface onto a 2-dimensional plane (**map**) with **constant units of distance (e.g., m), area, direction across the two dimensions**.

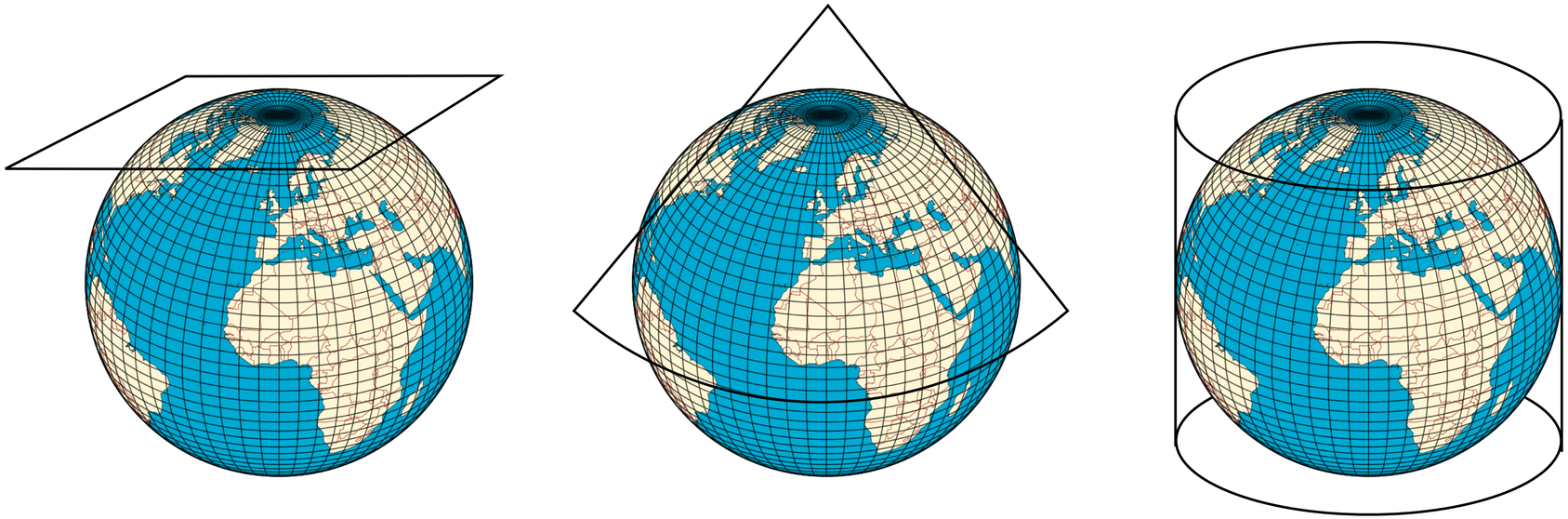
Crucially, **all map projections introduce distortion**. It's impossible to perfectly flatten a curved surface. Projections inevitably distort one or more properties.





Mercator is a cylindrical projected coordinate systems that **represents lines of constant angle** (i.e., course; good for nautical use).

Mercator projection, distortion increases towards higher latitudes.



Planar projection (poles)

Conical projection (mid-latitudes)

Cylindrical projection (rectangular maps)

There is no best projection. **Map-maker must decide which property is important** for the application (e.g., equal area for area calculations, equidistant for distance measurements).

What can GIS do in a SDM context?

Data Preparation & Cleaning: e.g., species occurrence records (vectors) mistakenly recorded out of their known distribution.

Environmental Extraction: Using the geographic coordinates of species occurrences to **extract the exact environmental values** (e.g., temperature, precipitation) from underlying continuous raster layers to build the modeling datasets.

Spatial Cropping & Masking: Trimming global environmental raster layers **to the specific study area**, focusing the model on accessible habitats.

Mapping & Visualization: Converting complex, continuous mathematical model outputs into **visual maps** - such as habitat suitability heatmaps or binary presence/absence maps - that can inform conservation and policy.